

Related Research

Annotated Bibliography

Scope note. Grouped by the modelling family each paper informs. Links marked (arXiv) were located and confirmed via live web search on 2026-08-28; a few foundational works are cited by author/venue only because this session could not re-verify a specific link.

1. Classical ML for network/KPI anomaly detection

- **"Machine Learning-Based Network Anomaly Detection: Design, Implementation, and Evaluation"**, MDPI Telecom 5(4):143, 2024. <https://www.mdpi.com/2673-2688/5/4/143> - Benchmarks tree-based and classical ML models (Isolation Forest, Random Forest, gradient boosting) for network KPI anomaly detection; useful as the "simple, auditable baseline" tier of the framework.
- **"A Unified Approach to Interpreting Model Predictions"**, Lundberg, S. & Lee, S. (SHAP), NeurIPS 2017. - The attribution method used throughout the framework's explainability component.

2. Deep learning for multivariate KPI time series

- **"Robust Anomaly Detection for Multivariate Time Series via Stochastic Recurrent Neural Network"**, Su, Y., Zhao, Y., Niu, C., Liu, R., Sun, W., Pei, D. (OmniAnomaly), KDD 2019. - Introduces the Server Machine Dataset (SMD) used as the public dataset in this notebook, and a VAE+RNN approach to multivariate KPI anomaly detection.
- **"Detecting Spacecraft Anomalies Using LSTMs and Nonparametric Dynamic Thresholding"**, Hundman, K. et al. (LSTM-NDT), KDD 2018. - Sequence-to-sequence LSTM forecasting plus dynamic thresholding for early-warning style anomaly flags, the direct ancestor of the "predict degradation before disruption" component.
- **"Graph Neural Network-Based Anomaly Detection in Multivariate Time Series"**, Deng, A. & Hooi, B. (GDN), AAAI 2021. - Combines a learned sensor-dependency graph with attention-based forecasting and per-sensor deviation scores, bridging the deep-learning and GNN families below.

3. Transformer-based approaches

- **"Transformers in Time Series: A Survey"**, Wen, Q. et al., arXiv:2202.07125. (arXiv) <https://arxiv.org/pdf/2202.07125> - Taxonomy of transformer variants (positional encoding, attention modules) for forecasting, anomaly detection, and classification; used to justify the encoder choice in Component 1.
- **"TranAD: Deep Transformer Networks for Anomaly Detection in Multivariate Time Series Data"**, Tuli, S., Casale, G., Jennings, N., VLDB 2022, arXiv:2201.07284. (arXiv) <https://arxiv.org/abs/2201.07284> - Transformer encoder with adversarial (two-phase) training and self-conditioning; strong reference for the forecasting/anomaly-scoring backbone.
- **"Root Cause Analysis of Anomalies in 5G RAN Using Graph Neural Network and Transformer"**, arXiv:2406.15638. (arXiv) <https://arxiv.org/html/2406.15638v1> - Directly telecom-RAN; combines a

transformer encoder with a GNN over cell/site topology for anomaly root-causing. This is the closest published analogue to the framework proposed here.

- **"Classification of Anomalies in Telecommunication Network KPI Time Series"**, arXiv:2308.16279. (arXiv) <https://arxiv.org/pdf/2308.16279> - Taxonomy of KPI anomaly types in real operator data (point, contextual, collective) that motivates the labeling scheme used in Component 1/2.
- **"Time Series Network Utilization KPI Forecasting Using Advanced AI/ML Models"**, arXiv:2607.19974. (arXiv) <https://arxiv.org/html/2607.19974> - Forecasting utilization-type KPIs with modern AI/ML models; supports the choice of probabilistic forecasting for the prediction component.

4. Graph Neural Networks for topology-aware root cause analysis

- **"MicroRCA: Root Cause Localization of Performance Issues in Microservices"**, Wu, L. et al., IFIP/IEEE IM 2020. - Builds an attributed service-dependency graph and uses Personalized PageRank to rank root-cause candidates; the pattern is reused with a cell/NE topology graph in Component 2.
- **"Cellular Network Fault Diagnosis Method Based on a Graph Convolutional Neural Network"**, PMC10459609. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10459609/> - GCN over cell-site topology for telecom fault diagnosis, with an LSTM front end for temporal priors.
- **"CHASE: A Causal Hypergraph based Framework for Root Cause Analysis in Multimodal Microservice Systems"**, arXiv:2406.19711. (arXiv) <https://arxiv.org/pdf/2406.19711> - Hypergraph extension for multimodal (metrics + logs + traces) causal root cause analysis.
- **"A Comprehensive Survey on Root Cause Analysis in (Micro) Services: Methodologies, Challenges, and Trends"**, arXiv:2408.00803. (arXiv) <https://arxiv.org/html/2408.00803v1> - Survey covering rule-based, ML, GNN, and causal RCA methods; used to cross-check that Component 2 covers the mainstream method families.

5. Causal reasoning: separating correlation from causal evidence

- **"Causation, Prediction, and Search"**, Spirtes, P., Glymour, C., Scheines, R. - Foundational source for the PC/FCI constraint-based causal discovery algorithms used in Component 3.
- **"Root Cause Analysis In Microservice Using Neural Granger Causal Discovery"**, RUN, arXiv:2402.01140. (arXiv) <https://arxiv.org/abs/2402.01140> - Neural Granger causal discovery plus contrastive learning to build a causal graph from monitoring metrics; direct template for the Granger-causality step used in this notebook.
- **"Root Cause Analysis for Microservices based on Causal Inference: How Far Are We?"**, arXiv:2408.13729. (arXiv) <https://arxiv.org/html/2408.13729v1> - Empirically compares PC, FCI, Granger, fGES, and LiNGAM as causal-graph builders for RCA; informs the choice and limits of Granger causality used here.
- **"Graphical Causal Reasoning for Root Cause Analysis in Cloud Networks"**, arXiv:2606.13532. (arXiv) <https://arxiv.org/html/2606.13532> - Constructs a causal graph from binary event time series via bivariate Granger causality plus conditional-independence tests, then walks the graph to rank causes - the same two-step (build graph -> rank) pattern as Component 2+3 here.

6. LLM-based approaches for diagnosis / operator-facing explanation

- **"Exploring LLM-based Agents for Root Cause Analysis"**, arXiv:2403.04123. (arXiv) <https://arxiv.org/pdf/2403.04123> - Autonomous LLM agents with tool use for cloud incident RCA (context management, trajectory stabilization); the design pattern used for the explanation-generation LLM in Component 4: tool-grounded, not free-generation.
- **"RCA Copilot: Transforming Network Data into Actionable Insights via Large Language Models"**, arXiv:2507.03224. (arXiv) <https://arxiv.org/pdf/2507.03224> - LLM pipeline that aggregates multimodal diagnostic evidence and generates root-cause narratives for network incidents; closest analogue to Component 4's report generator.
- **"Automatic Root Cause Analysis via Large Language Models for Cloud Incidents"**, arXiv:2305.15778. (arXiv) <https://arxiv.org/pdf/2305.15778> - LLM reads retrieved telemetry/runbooks to propose root causes; motivates keeping the LLM strictly as a narrator over verified evidence rather than an independent diagnoser, to reduce hallucinated causes.

7. Uncertainty quantification & trustworthy AI

- **"A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification"**, Angelopoulos, A. & Bates, S., arXiv:2107.07511. (arXiv) <https://arxiv.org/abs/2107.07511> - The conformal-prediction method used for Component 5: distribution-free coverage guarantees with no distributional assumptions on KPI noise.
- **"Conformal Prediction and Trustworthy AI"**, arXiv:2508.06885. (arXiv) <https://arxiv.org/abs/2508.06885> - Reviews why calibrated, guaranteed-coverage uncertainty (versus bare softmax confidence) is a prerequisite for trustworthy deployment.
- **"Uncertainty Aware Deep Learning Model for Secure and Trustworthy Channel Estimation in 5G Networks"**, arXiv:2305.02741. (arXiv) <https://arxiv.org/pdf/2305.02741> - A 5G-specific example of separating epistemic and aleatoric uncertainty in a deep model, mirroring the split used in Component 5.

Dataset

- **Su, Y. et al. (2019), Server Machine Dataset (SMD)**, <https://github.com/NetManAIOps/OmniAnomaly> - 28 machines x 38 metrics (CPU, memory, network, and load metrics sampled every minute), with anomaly-interval labels and ground-truth root-cause dimension labels (interpretation_label). Used in the notebook as a public, license-open proxy for telecom NE/CPU-memory/traffic KPI monitoring, chosen specifically because it is one of the few public datasets with root-cause ground truth. This enables quantitative validation of the root-cause component rather than only the anomaly-detection component. True RAN-level KPI datasets (latency, jitter, handover-failure with root-cause labels) are typically held by operators/vendors and are not public.

Framework-to-Literature Mapping

Framework Area	Purpose	Key Literature Families
Component 1	Prediction & anomaly detection	Classical ML, LSTM/VAE, Transformer
Component 2	Topology-aware root cause analysis	GDN, MicroRCA, GCN, 5G GNN + Transformer
Component 3	Causal validation	PC/FCI, Granger causality, causal graph ranking
Component 4	Operator-facing explanation	Tool-grounded LLM agents, RCA Copilot
Component 5	Uncertainty & trust	Conformal prediction, epistemic/aleatoric uncertainty